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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

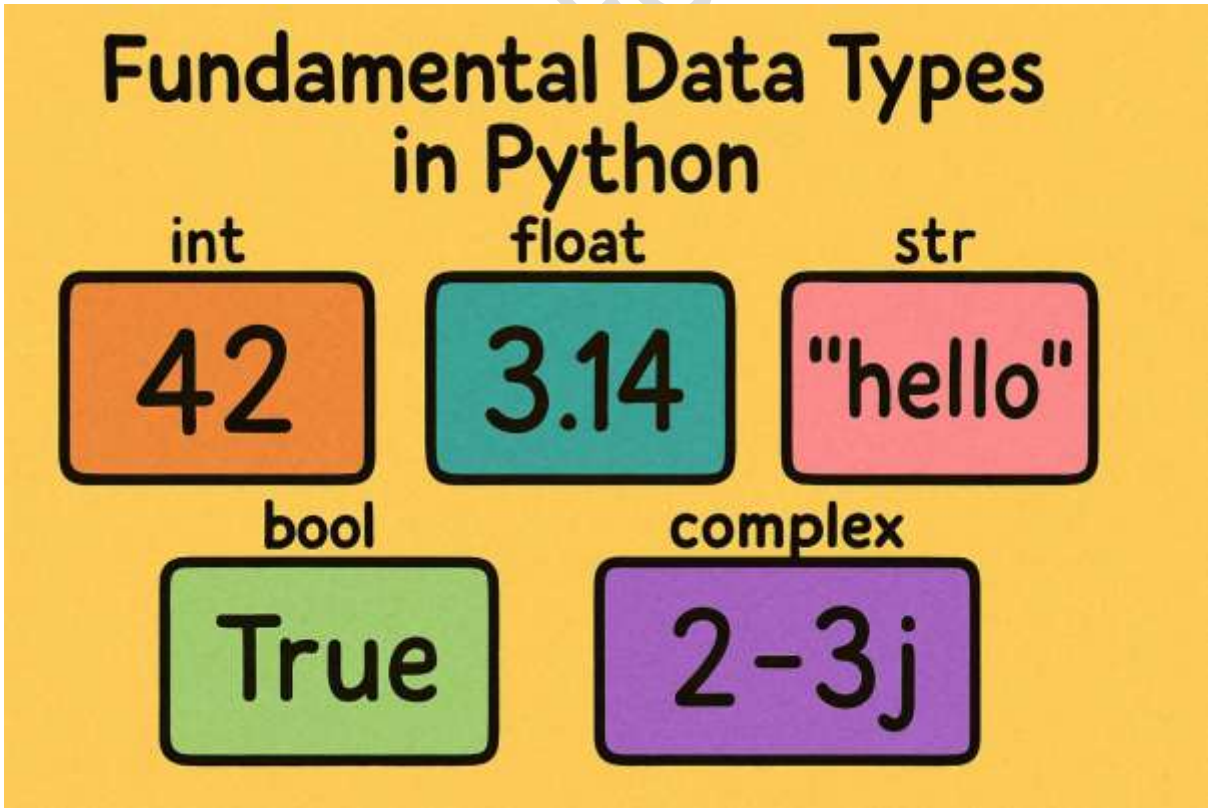
Daily Teaching : 2 Hours

Total Classes : 100

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Level 4 – AI Professional Foundation(11th–12th Standard)

Class 3 – Advanced Variables & Data Types



Learning Objectives

By the end of this class, students will be able to:

- Understand variables in Python.
- Identify different Python data types.
- Understand dynamic typing.
- Use type() to identify data types.
- Convert values from one data type to another.
- Understand int(), float(), str() and bool().
- Perform type conversion safely.
- Identify common conversion errors.
- Build practical programs using type conversion.

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1. What is a Variable?

A **variable** is a name used to store a value in a Python program.

For example:

student_name = "Aisha"

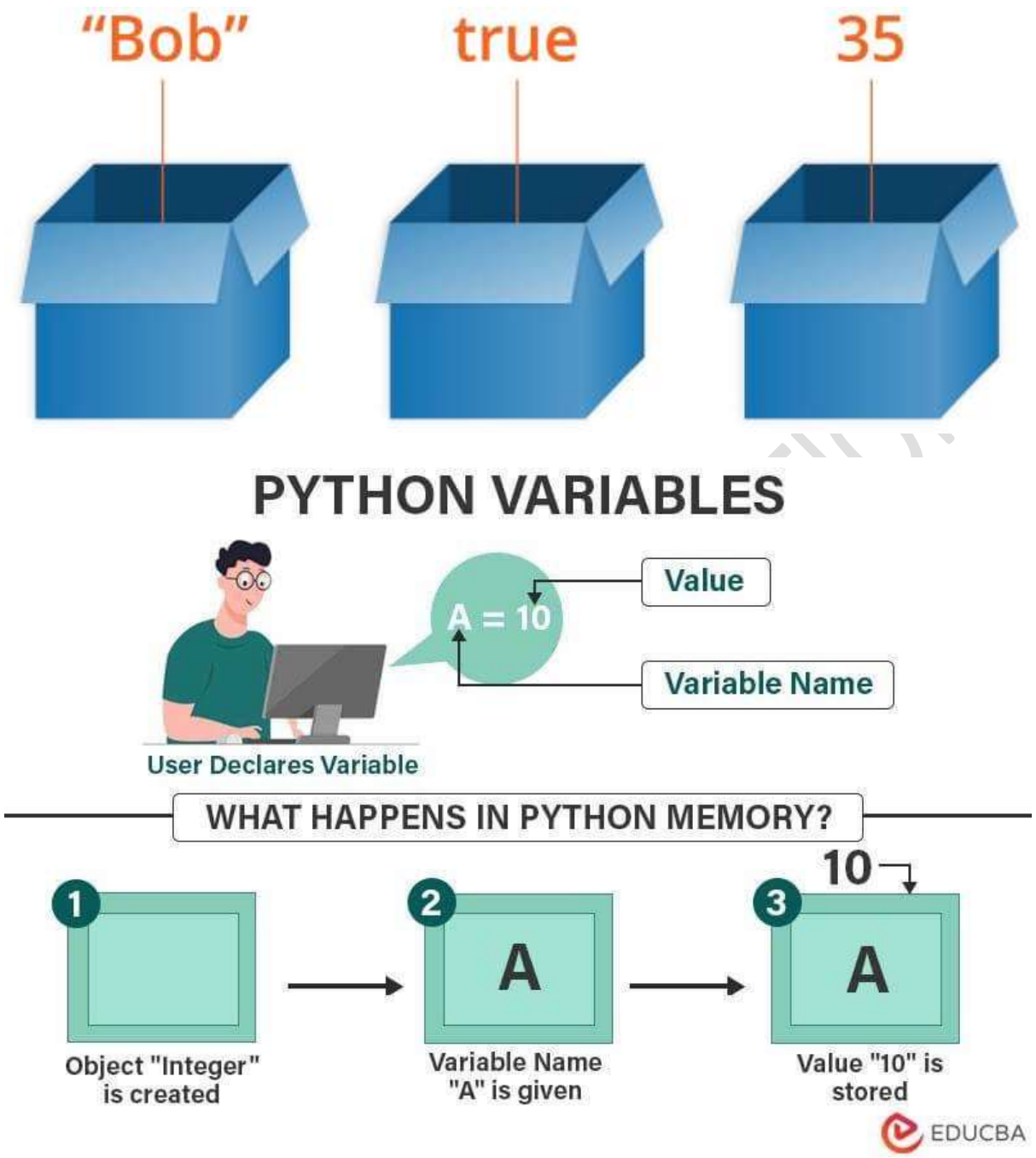
age = 17

marks = 92.5

Here:

- student_name stores text.
- age stores an integer.
- marks stores a decimal number.

Think of a variable as a **label attached to a box containing information**.



Example

name = "Rahul"

We can imagine:

name ——————▶ "Rahul"

The variable name refers to the value "Rahul".

2. Python is Dynamically Typed

Python does not require us to declare the data type before creating a variable.

Example:

value = 100

Python understands that value is an integer.

Later, the same variable can refer to another type:

value = "Python"

Now value refers to a string.

This is called **dynamic typing**.

3. Python Data Types

Python provides several built-in data types.

Data Type	Example	Description
int	25	Whole numbers
float	25.5	Decimal numbers
str	"Hello"	Text
bool	True	True/False
list	[10, 20, 30]	Ordered collection
tuple	(10, 20, 30)	Ordered immutable collection
set	{10, 20, 30}	Unique values
dict	{"name": "Aisha"}	Key-value data

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Data Type Example	Description
NoneType None	No value

4. Integer – int

An integer is a whole number without a decimal point.

age = 17

students = 45

temperature = -5

Examples:

10

25

0

-15

1000

5. Float – float

A float represents a number containing a decimal part.

percentage = 87.5

price = 499.99

height = 5.8

Examples:

10.5

25.75

0.5

-3.14

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6. String – str

A string is a sequence of characters.

name = "Aisha"

city = "Hyderabad"

course = "Advanced Python"

Strings can contain:

- Letters
- Numbers
- Spaces
- Symbols

Example:

student_id = "1025"

Although "1025" contains digits, it is a **string**, not an integer, because it is inside quotation marks.

7. Boolean – bool

Boolean values represent two possibilities:

True

False

Example:

is_student = True

is_logged_in = False

Booleans are commonly used in conditions.

age = 17

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```
is_adult = age >= 18


print(is_adult)
```

Output:

False

8. Finding the Data Type with type()

Python provides the type() function.

```
age = 17

print(type(age))
```

Output:

<class 'int'>

Another example:

```
name = "Aisha"

marks = 92.5

passed = True


print(type(name))

print(type(marks))

print(type(passed))
```

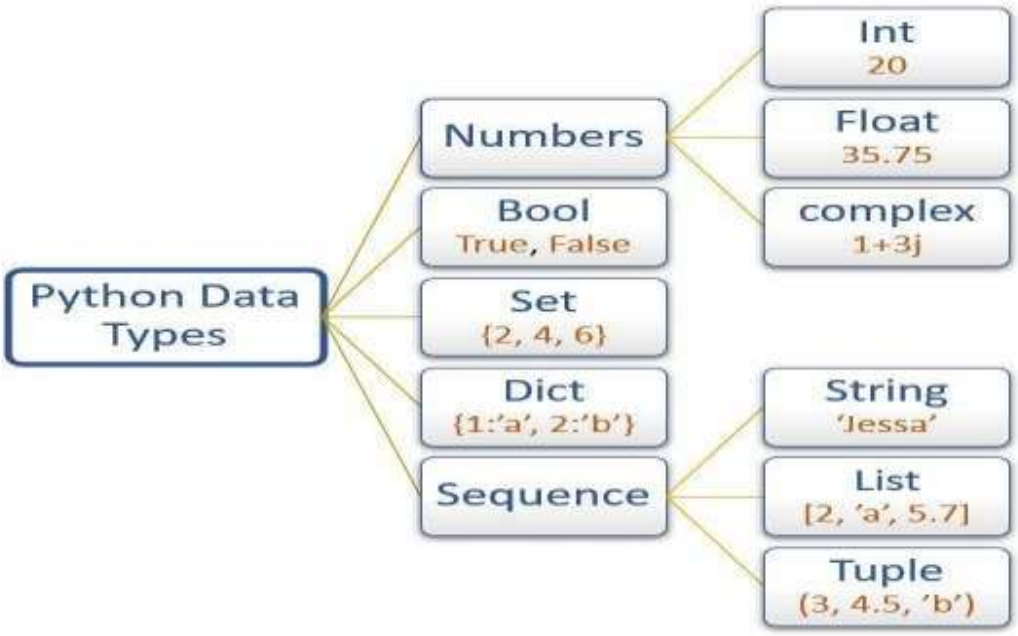
Output:

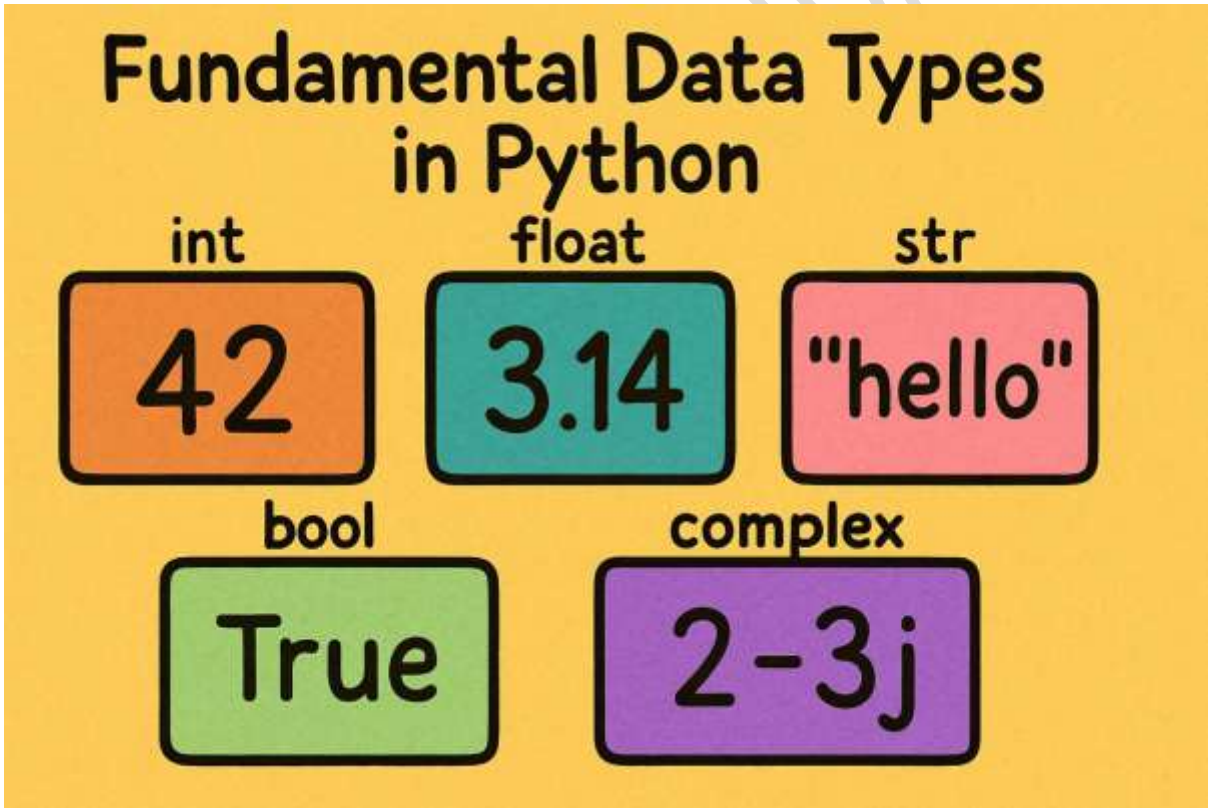
<class 'str'>

<class 'float'>

<class 'bool'>

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Python Data Types
Numbers
Int
20
Float
35.75
complex
1+3j
Bool
True, False
Set
{2, 4, 6}
Dict
{1:'a', 2:'b'}
Sequence
String
'Jessa'
List
[2, 'a', 5.7]
Tuple
(3, 4.5, 'b')


Fundamental Data Types in Python
int
42
float
3.14
str
"hello"
bool
True
complex
2-3j

Data Type	Key Characteristics	Example Code	Output
Integer (int)	- Whole numbers (no decimals). - Can be positive or negative. - Unlimited length.	x = 42 y = -100 z = 12345678901234567890	42 -100 12345678901234567890
Float (float)	- Numbers with decimal points. - Can be in exponential form (e.g., 2.5e2). - Limited precision.	a = 3.14 b = -0.001 c = 2.5e2	3.14 -0.001 250.0
String (str)	- Enclosed in single (') or double (") quotes. - Immutable (cannot be changed after creation). - Supports slicing and indexing.	s1 = "Hello" s2 = 'Python' s3 = "123"	"Hello" 'Python' "123"
Boolean (bool)	- Only two values: True or False. - Used in logical operations and conditions. - Non-zero numbers and non-empty strings are True.	b1 = True b2 = False b3 = bool(0) b4 = bool("Hello")	True False False True

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9. What is Type Conversion?

Type conversion means changing a value from one data type to another.

For example:

String → Integer

Integer → Float

Integer → String

Float → Integer

String → Float

Python provides built-in functions to perform many conversions.

10. Converting String to Integer

Use:

int()

Example:

age = "17"

age = int(age)

print(age)

print(type(age))

Output:

17

<class 'int'>

Why is this important?

input() returns a string.

For example:

age = input("Enter your age: ")

Even if the student enters:

17

Python initially stores it as:

"17"

To perform numerical calculations:

age = int(input("Enter your age: "))

11. String to Float

Use:

float()

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Example:

price = "499.50"

price = float(price)

print(price)

print(type(price))

Output:

499.5

<class 'float'>

12. Integer to String

Use:

str()

Example:

age = 17

age_text = str(age)

print(age_text)

print(type(age_text))

Now the number 17 has become the string "17".

13. Integer to Float

number = 25

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```
decimal_number = float(number)


print(decimal_number)
```

Output:

25.0

14. Float to Integer

```
number = 25.9


whole_number = int(number)


print(whole_number)
```

Output:

25

Important

Converting a float to an integer **removes the decimal part**. It does not round normally.

For example:

```
print(int(9.99))
```

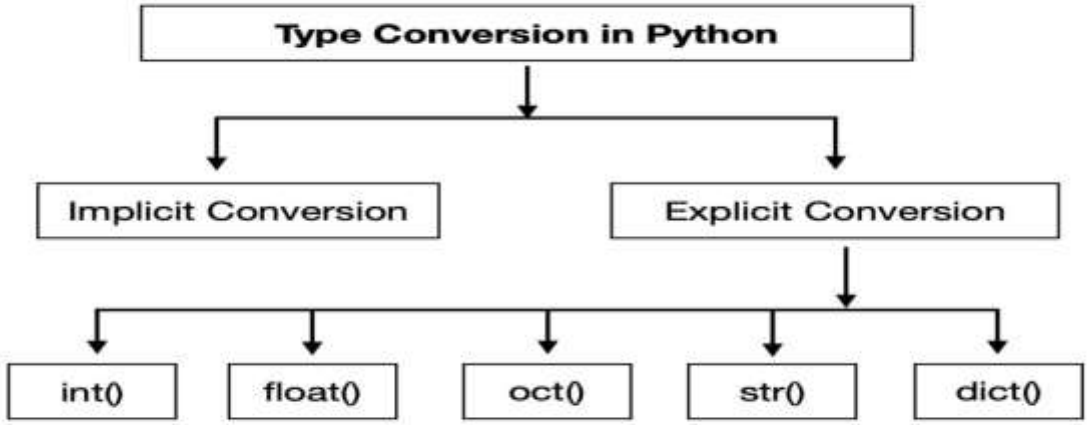
Output:

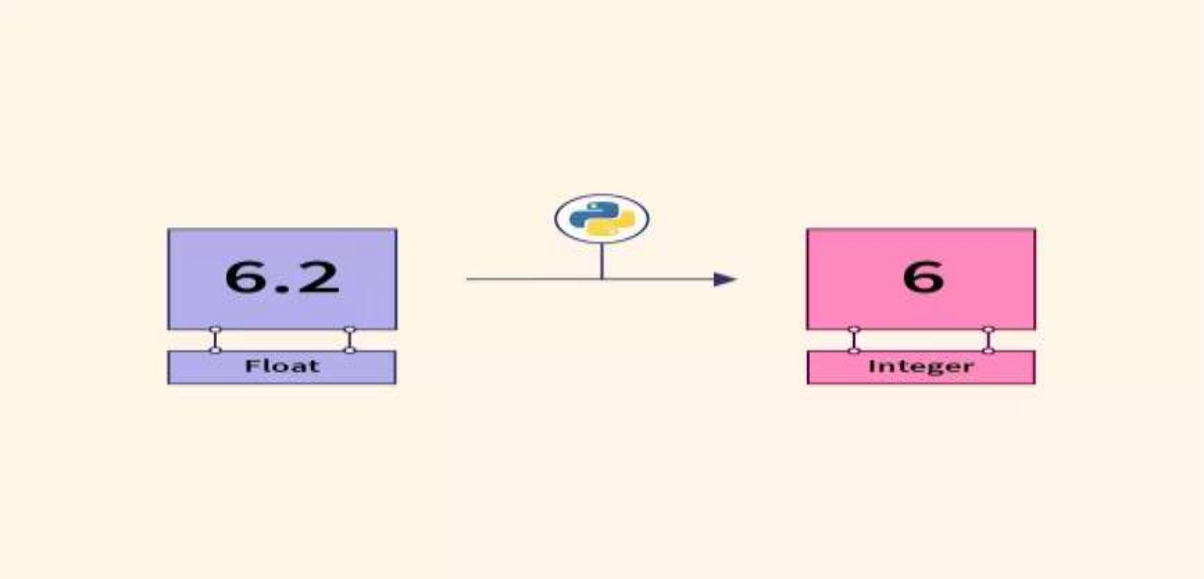
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15. Type Conversion Map

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Think of conversions like this:

"25"

|

int()

↓

25

|

float()

↓

25.0

|

str()

↓

"25.0"

16. Type Conversion Errors

Not every string can be converted to an integer.

This works:

number = int("25")

But this does not:

number = int("hello")

Python will generate a:

ValueError

Similarly:

price = float("abc")

will cause a conversion error.

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Students will learn how to handle these errors properly in the upcoming **Exception Handling** lessons.

17. Why input() and Type Conversion Matter

Consider:

num1 = input("Enter first number: ")

num2 = input("Enter second number: ")

print(num1 + num2)

If the student enters:

10

20

The output will be:

1020

Why?

Because both values are strings.

Correct Program

num1 = int(input("Enter first number: "))

num2 = int(input("Enter second number: "))

print(num1 + num2)

Output:30

Practical Session

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18. Practical Activity 1 – Student Age Calculator

Ask the student to enter their birth year.

```
birth_year = int(input("Enter your birth year: "))

current_year = 2026


age = current_year - birth_year


print("Your age is:", age)
```

Example

Enter your birth year: 2009

Your age is: 17

Challenge

Ask students to modify the program to calculate the approximate age after 5 years.

19. Practical Activity 2 – Marks Calculator

```
maths = float(input("Enter Maths marks: "))

science = float(input("Enter Science marks: "))

python = float(input("Enter Python marks: "))


total = maths + science + python

average = total / 3


print("Total:", total)

print("Average:", average)
```

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Example

Enter Maths marks: 90

Enter Science marks: 85

Enter Python marks: 95

Total: 270

Average: 90.0

20. Practical Activity 3 – Shopping Bill

```
price = float(input("Enter product price: "))
quantity = int(input("Enter quantity: "))

total = price * quantity

print("Total amount:", total)
```

Example

Enter product price: 250.50

Enter quantity: 3

Total amount: 751.5

21. Practical Activity 4 – Temperature Conversion

Students can convert Celsius to Fahrenheit.

Formula:

Fahrenheit = (Celsius × 9/5) + 32

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Python program:

```
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9 / 5) + 32

print("Temperature in Fahrenheit:", fahrenheit)
```

Example

Enter temperature in Celsius: 25

Temperature in Fahrenheit: 77.0

22. Practical Activity 5 – Currency Calculation

Create a simple currency calculation program.

```
amount = float(input("Enter amount: "))

rate = float(input("Enter conversion rate: "))

converted_amount = amount * rate

print("Converted amount:", converted_amount)
```

Learning Point

This teaches students how to accept decimal values and perform calculations after converting input.

23. Practical Activity 6 – Data Type Detective

Give students this program:

```
name = "Rahul"
```

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```
age = 17

marks = 88.5

passed = True


print(type(name))

print(type(age))

print(type(marks))

print(type(passed))
```

Students predict the output before running it.

Expected:

```
<class 'str'>

<class 'int'>

<class 'float'>

<class 'bool'>
```

24. Interactive Challenge – Fix the Program

Give students:

```
num1 = input("Enter first number: ")

num2 = input("Enter second number: ")


result = num1 + num2


print("Result:", result)
```

Question

If the user enters 15 and 25, why does the program produce:

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1525

instead of:

40

Student Task

Fix the program.

Correct Answer

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

result = num1 + num2

print("Result:", result)
```

25. Classroom Game – Type Conversion Race

Divide the class into teams.

Give each team the following:

"100"

25

25.5

True

"50.75"

Students must identify:

1. Original data type.
2. Which conversion can be performed.
3. Result after conversion.

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Example

Value	Original Type	Conversion	Result
"100"	str	int()	100
25	int	float()	25.0
25.5	float	int()	25
25	int	str()	"25"
"50.75"	str	float()	50.75

Quick Quiz

Q1. Which function tells us the data type?

Answer: type()

Q2. Which function converts a value to an integer?

Answer: int()

Q3. Which function converts a value to a decimal number?

Answer: float()

Q4. Which function converts a value to a string?

Answer: str()

Q5. What type does input() return?

Answer: str by default.

Q6. What is the result?

int(15.9)

Answer: 15

Q7. What happens here?

int("Python")

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Answer: A ValueError occurs.

27. Mini Project – Student Marks Analyzer

Students should create a complete program.

Requirements

The program should:

1. Ask for student name.
2. Ask for marks in three subjects.
3. Convert the marks to numbers.
4. Calculate total.
5. Calculate average.
6. Display the student's information.

Solution

```
student_name = input("Enter student name: ")


maths_marks = float(input("Enter Maths marks: "))
science_marks = float(input("Enter Science marks: "))
python_marks = float(input("Enter Python marks: "))


total_marks = maths_marks + science_marks + python_marks
average_marks = total_marks / 3


print("\n--- Student Result ---")
print("Name:", student_name)
print("Total Marks:", total_marks)
```

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```
print("Average Marks:", average_marks)
```

Homework

Basic

Write programs for:

1. Convert "100" into an integer.
2. Convert "45.5" into a float.
3. Convert 500 into a string.
4. Convert 20 into a float.
5. Display the type of five different variables.

Intermediate

Create a program that accepts:

Student Name

Age

Height

Marks

Convert each value to the appropriate data type and display the result.

Advanced

Create a **Student Result Calculator** that:

- Accepts five subject marks.
- Converts input to float.
- Calculates total and average.
- Displays the data type of each mark.
- Determines whether the student passed.

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2-Hour Class Plan

Time	Activity
0–10 min	Revision of Class 2
10–25 min	Variables
25–45 min	Python data types
45–60 min	type() function
60–80 min	Type conversion
80–100 min	Practical exercises
100–110 min	Type Conversion Race
110–115 min	Quiz
115–120 min	Homework & recap

Student Notes – Remember

int() → Integer

float() → Decimal number

str() → String

bool() → Boolean

type() → Check data type

Most Important Concept

age = input("Enter age: ")

input() gives a **string**.

For numerical calculations:

age = int(input("Enter age: "))

For decimal calculations:

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```
marks = float(input("Enter marks: "))
```

Key Message

Understanding data types and type conversion is essential because real-world programs constantly receive, process and transform different kinds of data.

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